

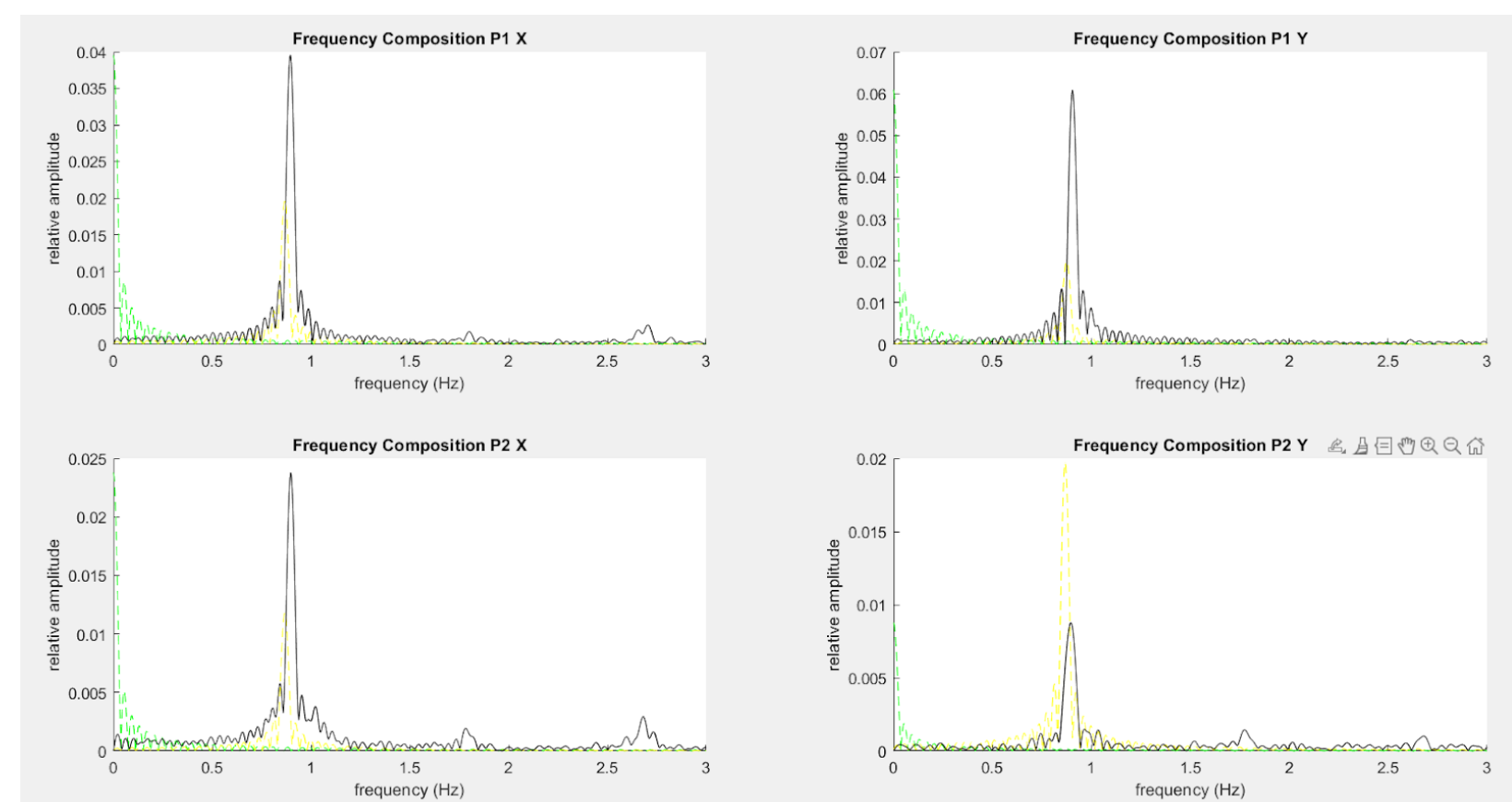
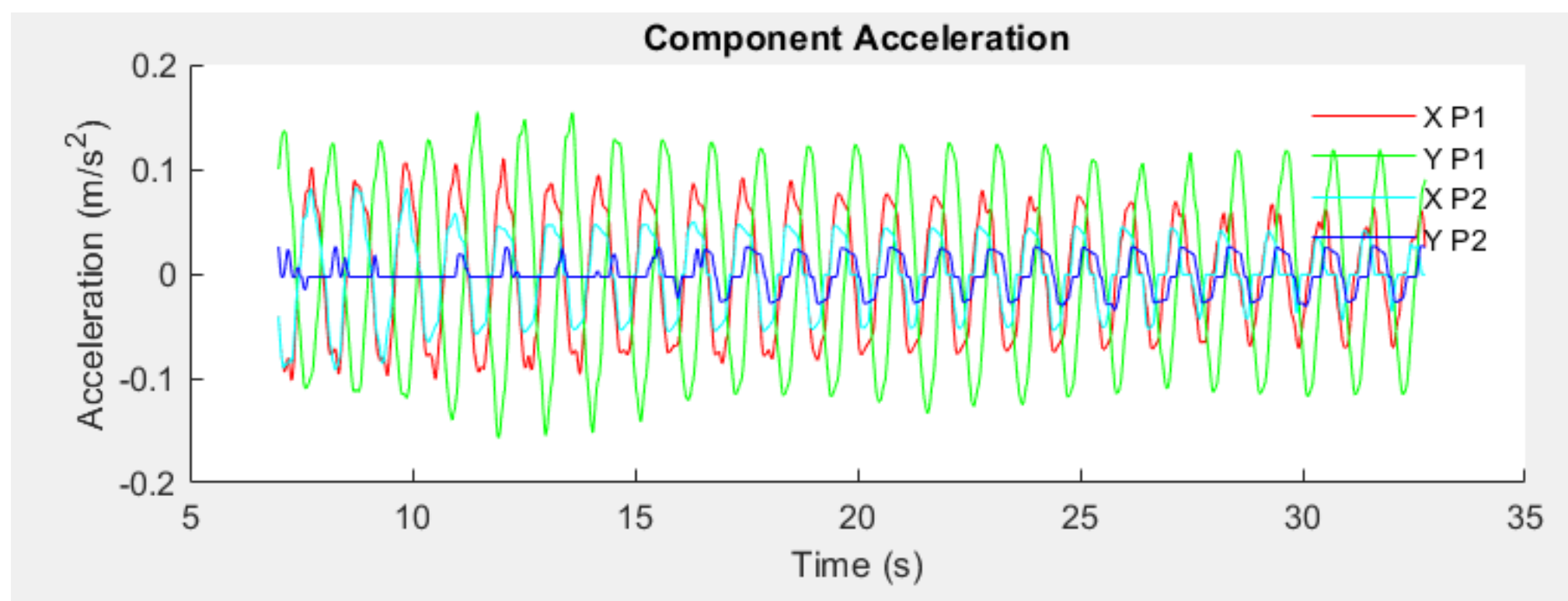
Objective

- Build a square array of coupled pendula** with tunable coupling, allowing motion in two orthogonal degrees of freedom.
- Identify and measure the system's normal modes** using controlled initial conditions and accelerometer data and compare observed frequencies with theoretical predictions.
- Examine the role of symmetry** in shaping mode structure and frequency composition, including effects of intentional or accidental symmetry breaking.

Background and Motivation

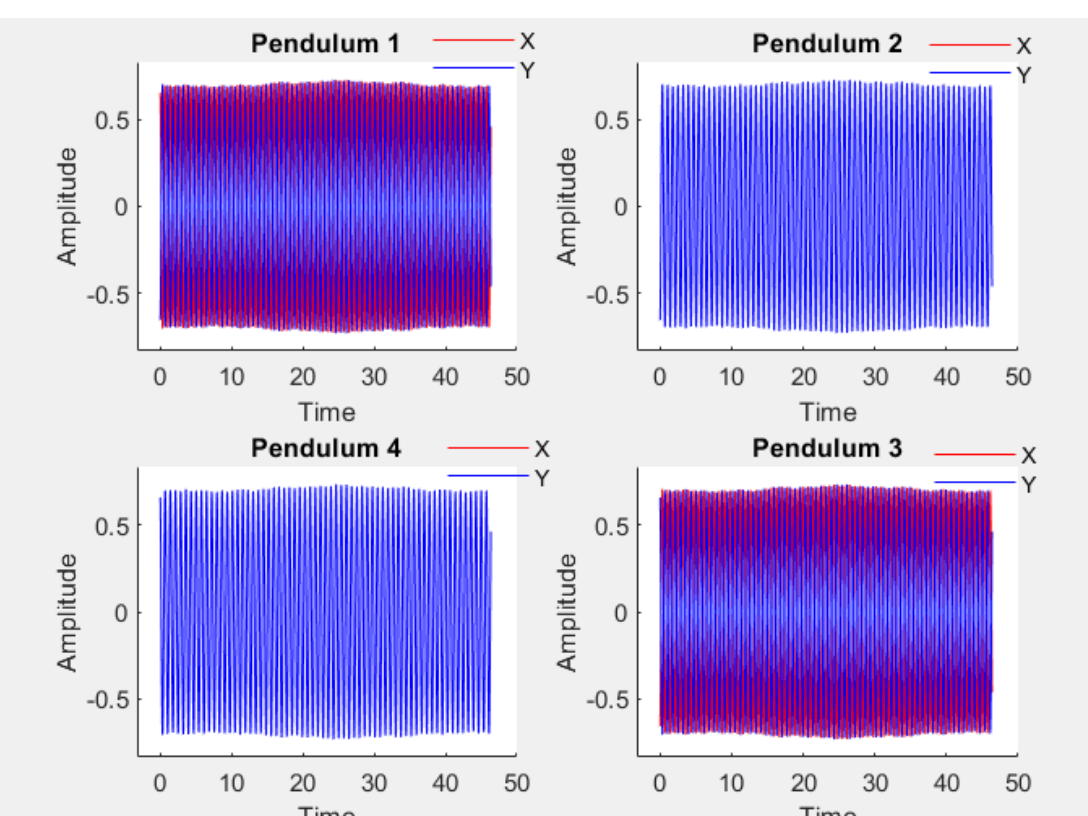
In the first lab of the semester, we studied the normal modes of a pair of coupled pendulums. This motivated our project: extending these principles to a higher-dimensional system to examine how additional degrees of freedom affect the structure of the normal modes. By moving to a two-dimensional array, we aimed to observe how the added DOFs generate new collective behaviors not present in the basic two-pendulum setup.

For the first data points, we chose to study the symmetric and anti-symmetric modes that serve as the natural extensions of the 1-DOF normal modes (see figures below). The frequency composition of these motions acted as a reference when identifying normal modes, which we defined as oscillations where all components of the system move at a single shared frequency, indicating a coherent collective motion of the coupled pendula.

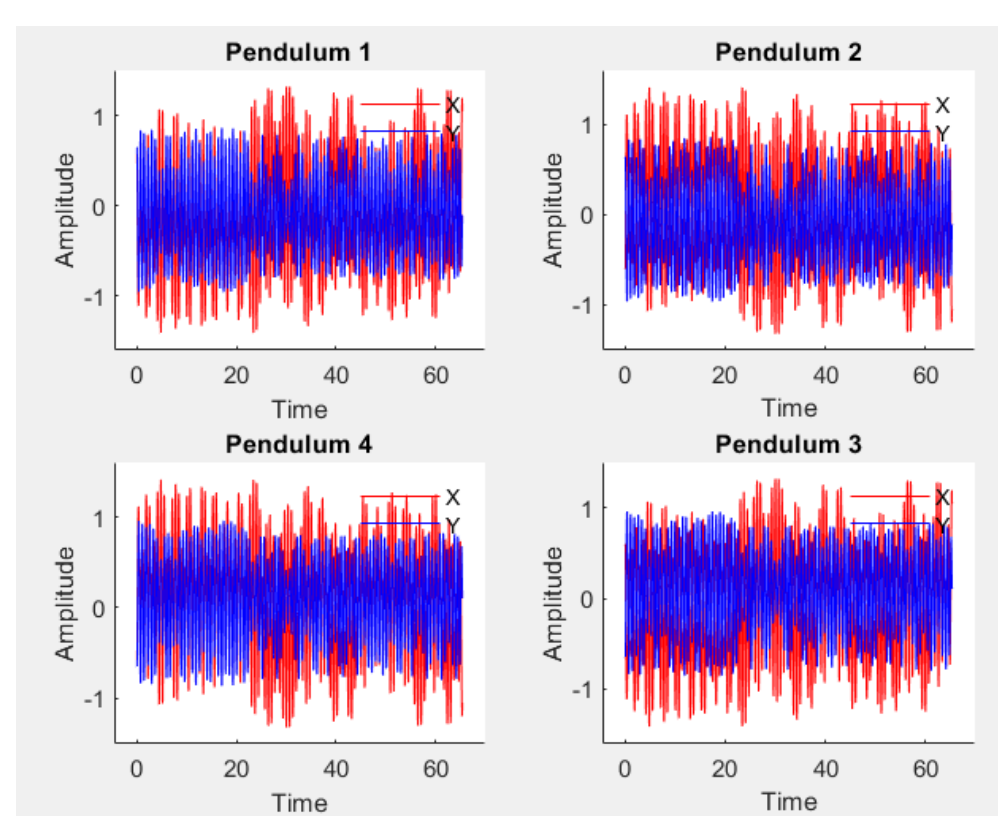


Analysis– Simulated Results

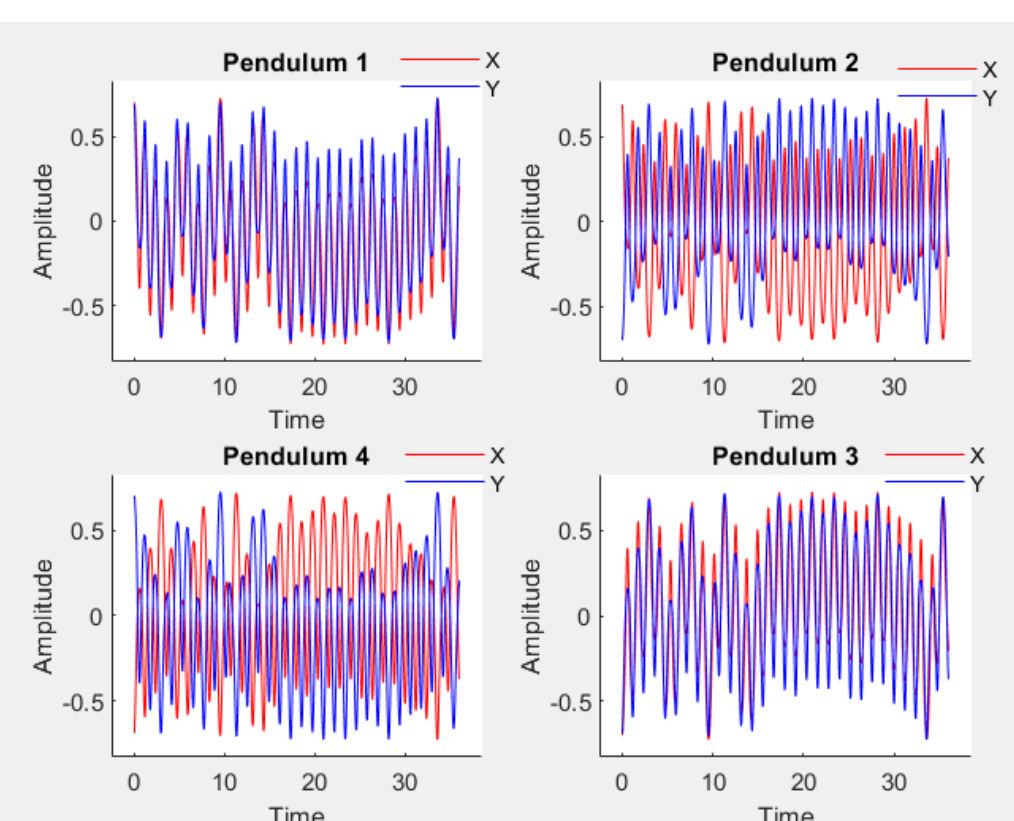
To supplement lab data, a simulation was programmed (red and blue data) to demonstrate the position of ideal pendula as a function of time. These simulations demonstrate that, even under ideal circumstances, (4) anti-symmetric breathing and (10) circle modes are indeed not normal. They also show what the position as a function of time would look like for a normal mode (including "assisted parallel rotating (11)").



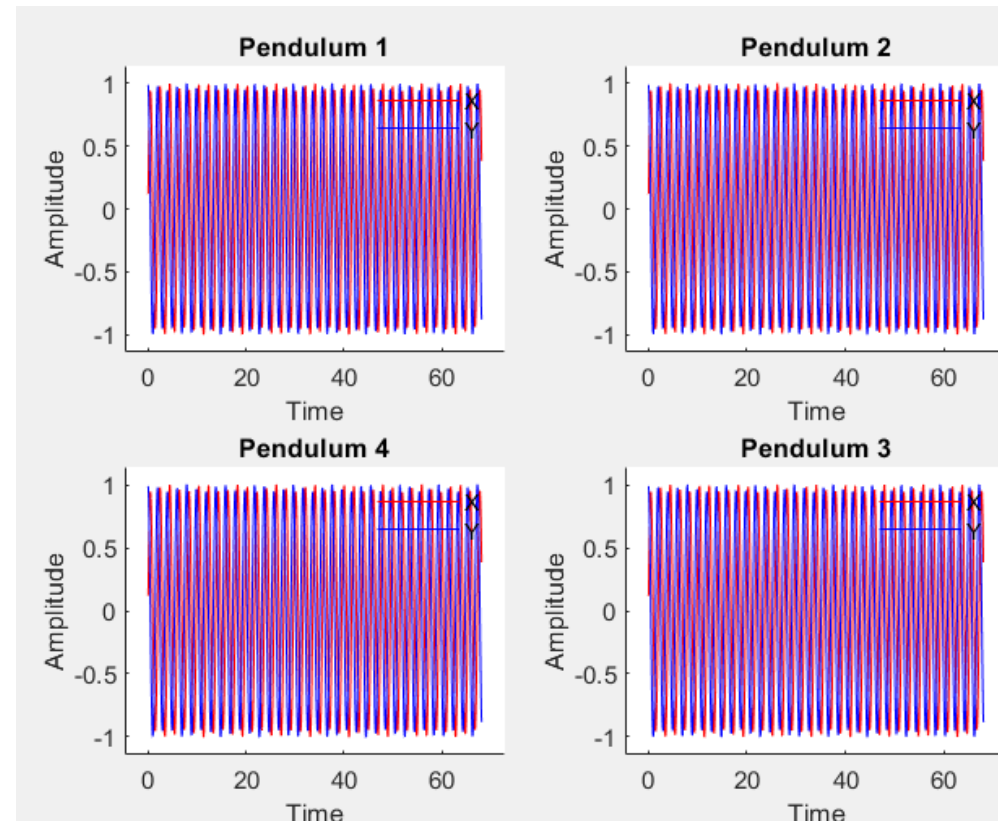
Breathing (3)



Trapezoid (8)



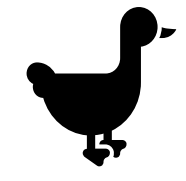
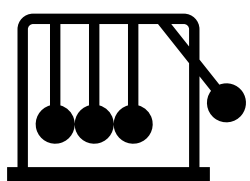
Circle (10)



Assisted Parallel Rotating (11)

Exploring the Normal Modes of Coupled Pendula in 2D

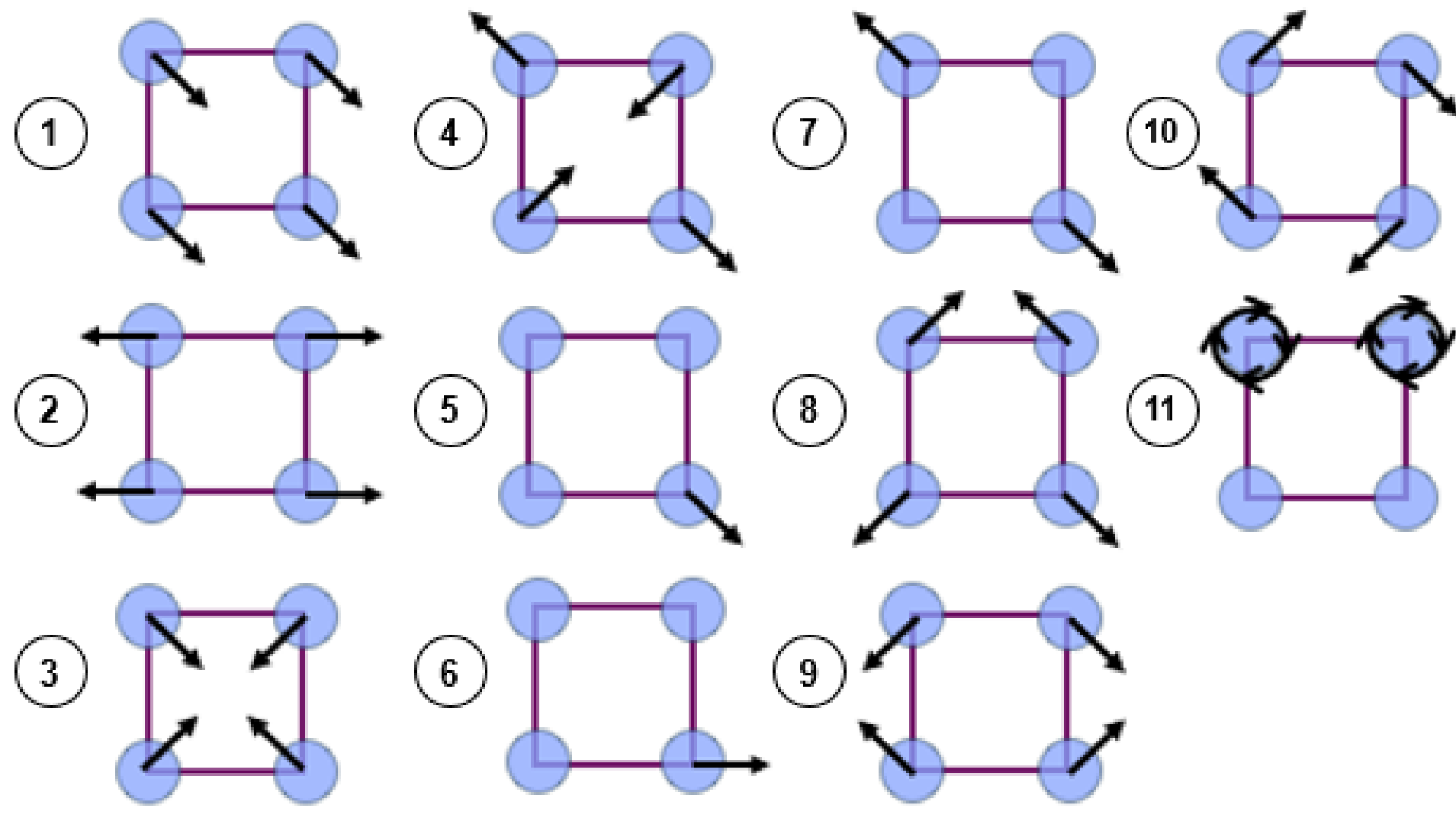
Anna Burgess, Hannah Murphy
Physics 15 C Lab Final Project



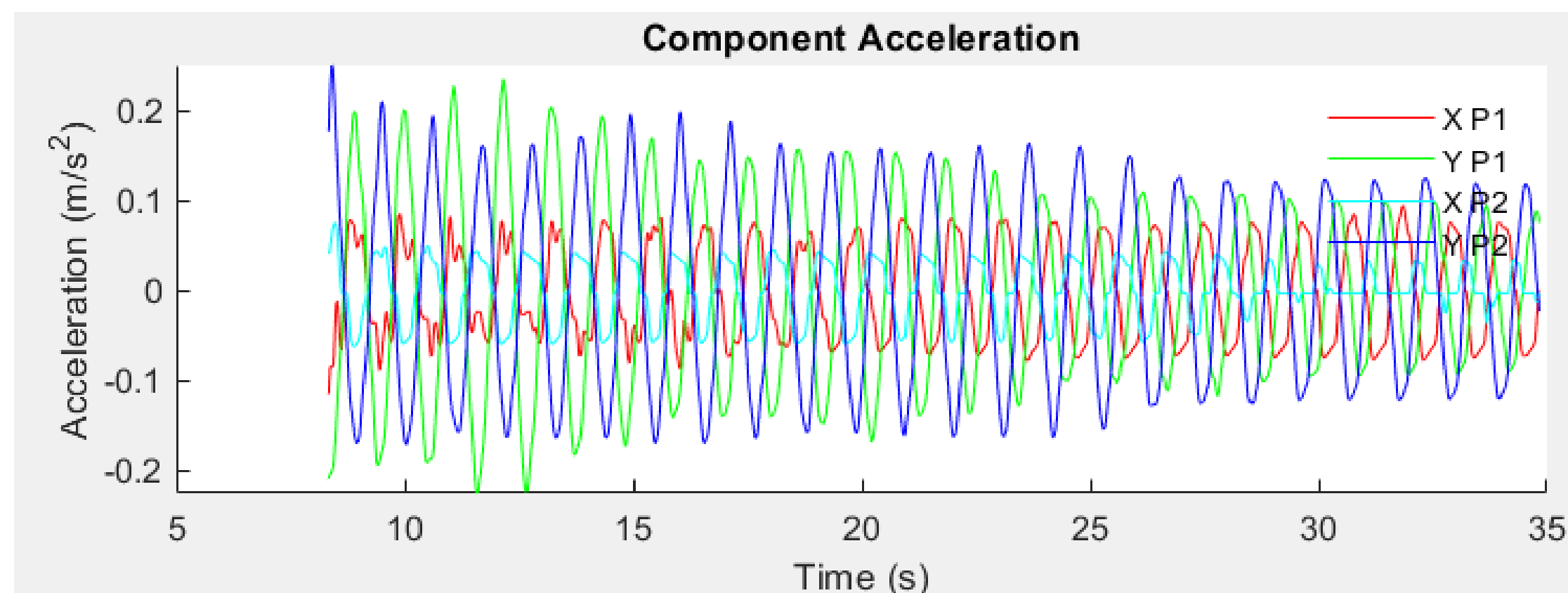
Data

For each of our 11 displacement schemes, we recorded 25 seconds of data at a 115200 baud output rate with a 500 Hz internal sampling rate. To calibrate the accelerometers, the system was left completely at rest for the first two seconds of each run, allowing the Arduino to establish baseline readings and zero the sensors before motion began. This ensured that all subsequent measurements reflected only dynamic acceleration from the coupled pendulum motion.

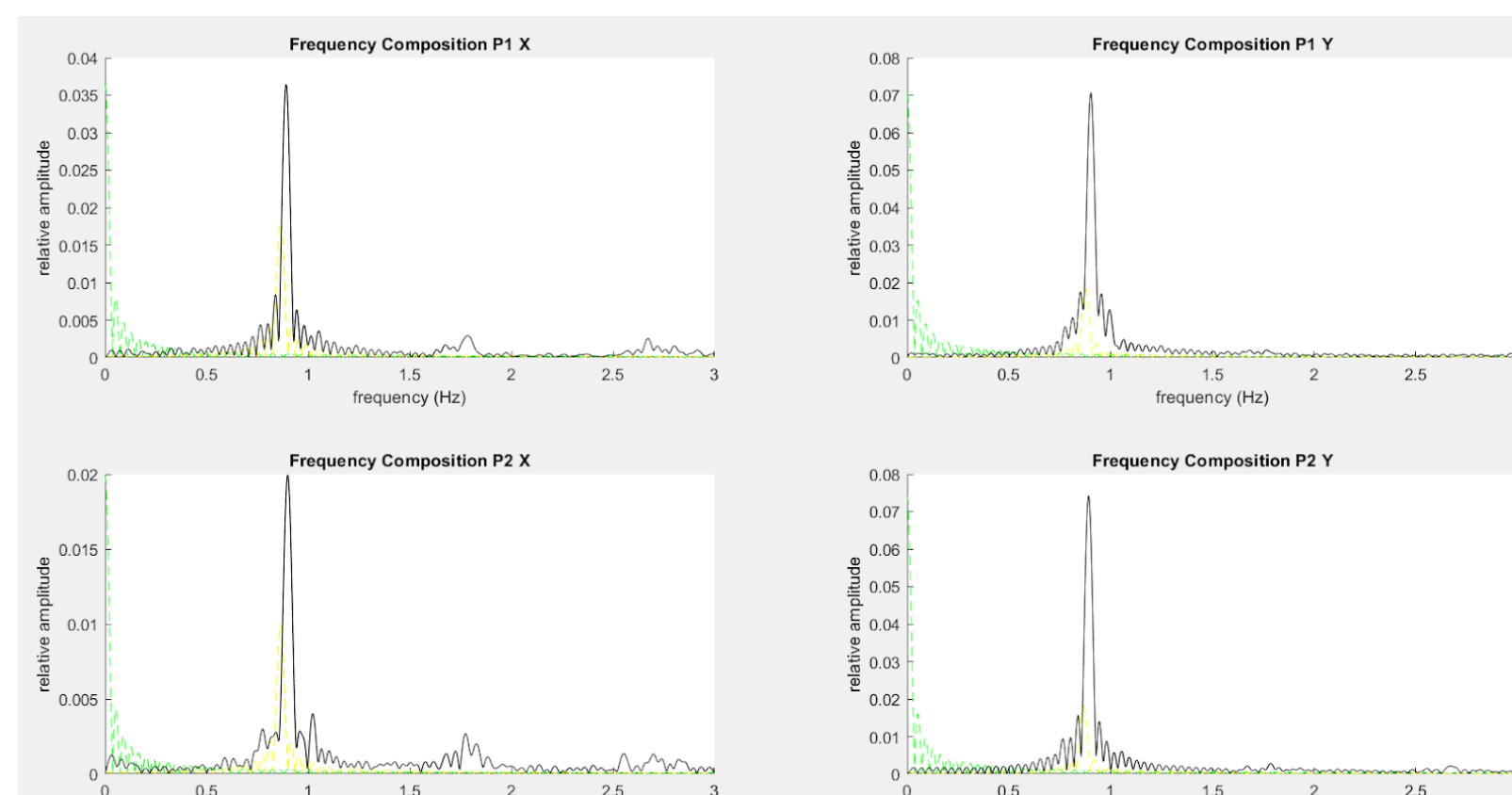
No.	Name	Normal Mode?
1	Symmetric	Yes
2	Anti-Symmetric	Yes
3	Breathing	Yes
4	Anti-Symmetric Breathing	No
5	Singular Radial	No
6	Singular Linear	No
7	Dual Radial (Opposite Corner)	No
8	Trapezoid	No
9	Rectangle	Yes
10	Circle	No
11	Assisted Parallel Rotating	Yes



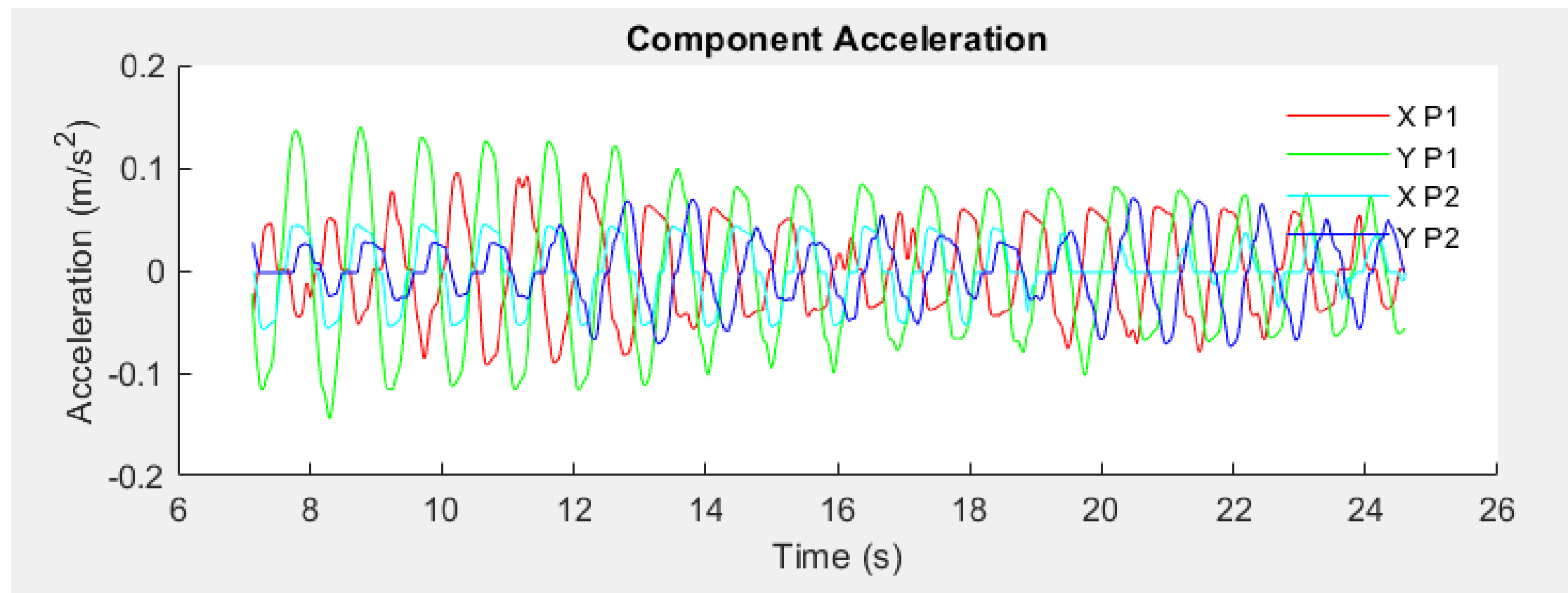
Analysis– Experimental Results



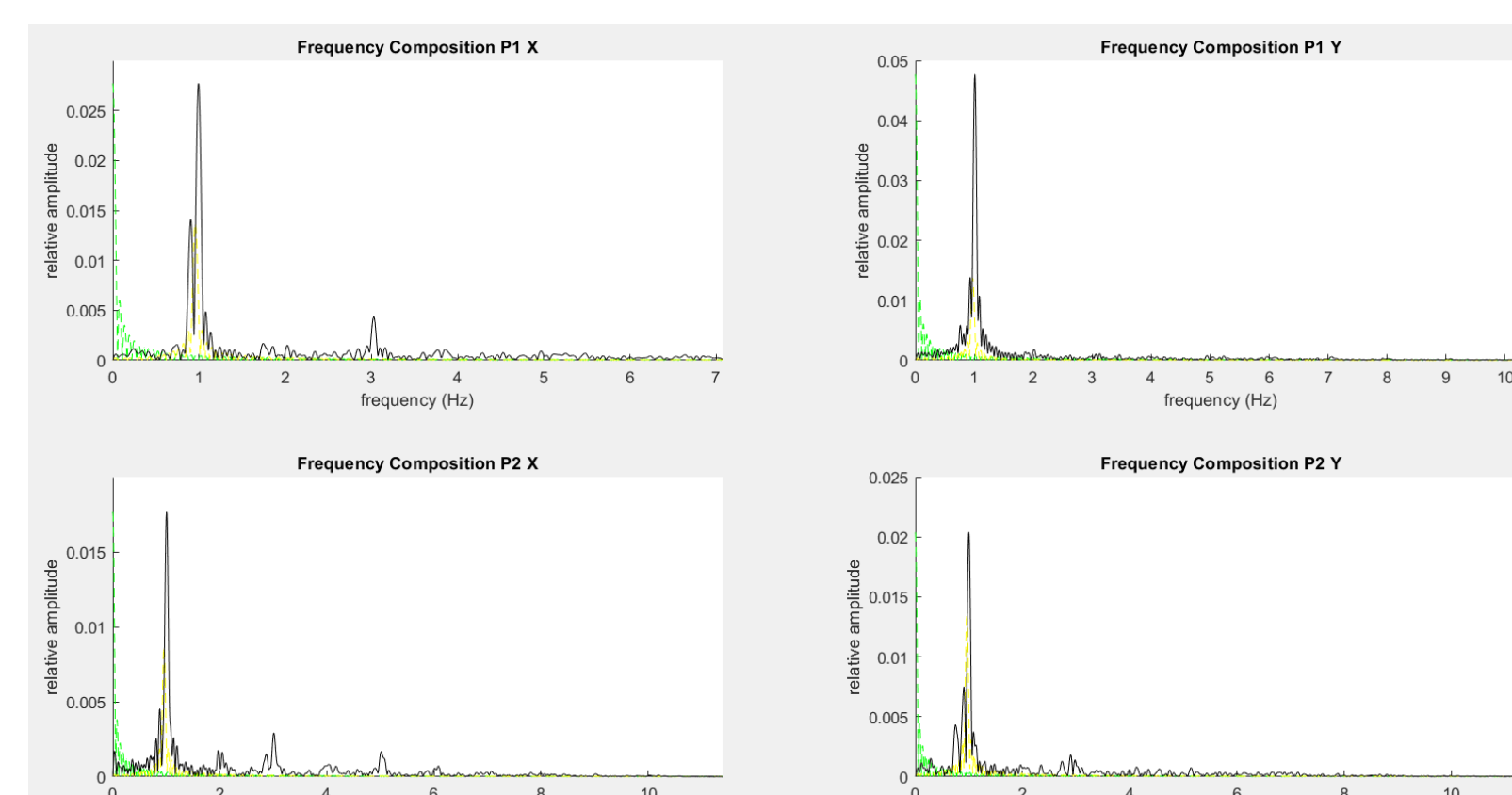
Acceleration in X and Y for Breathing Mode (3)



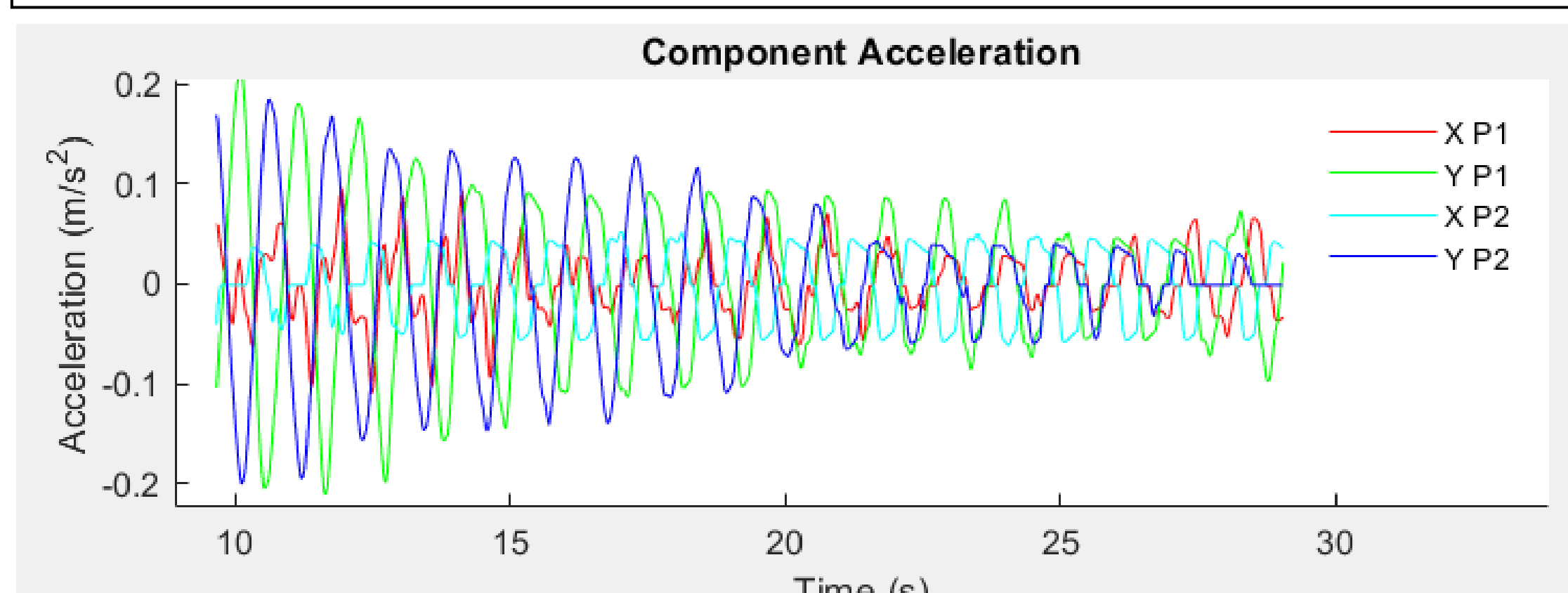
Frequency Components for each Axis



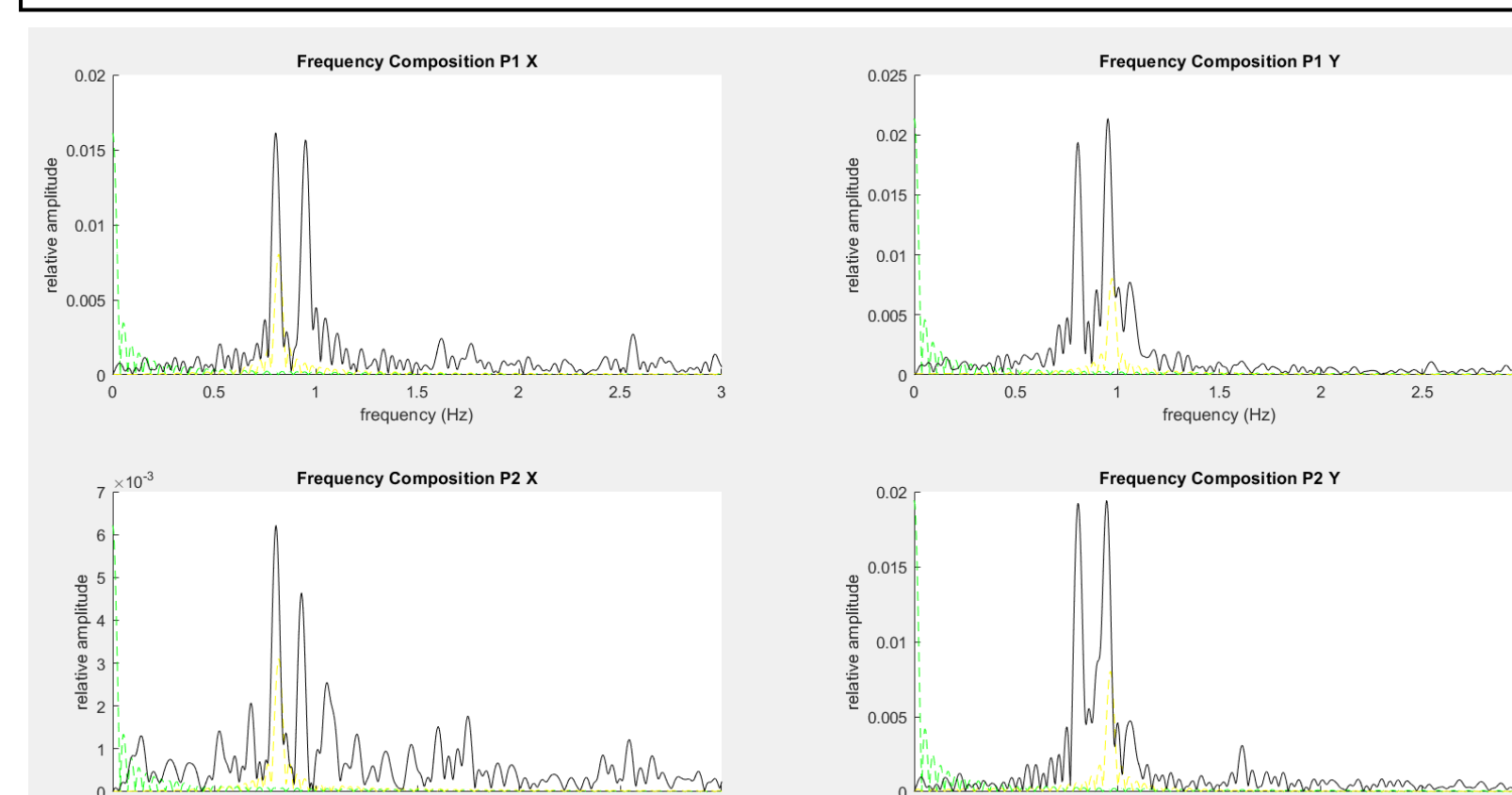
Acceleration in X and Y for Trapezoid Mode (8)



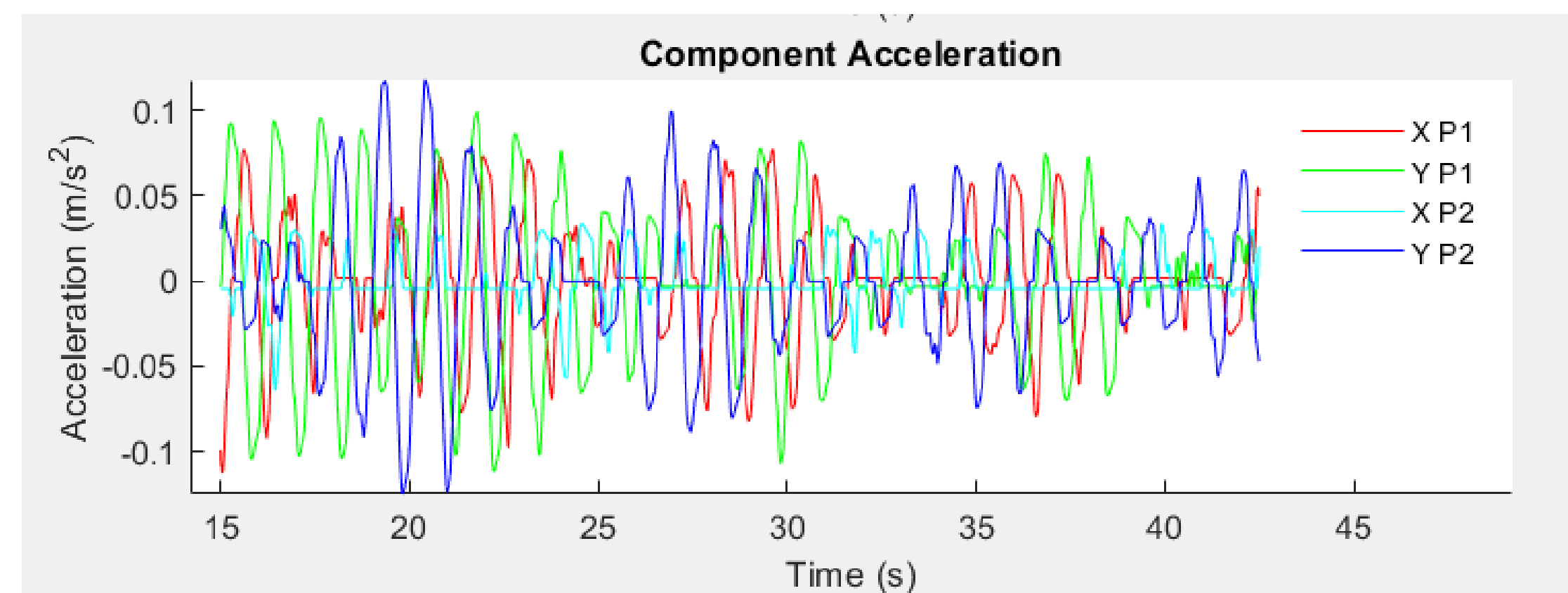
Frequency Components for each Axis



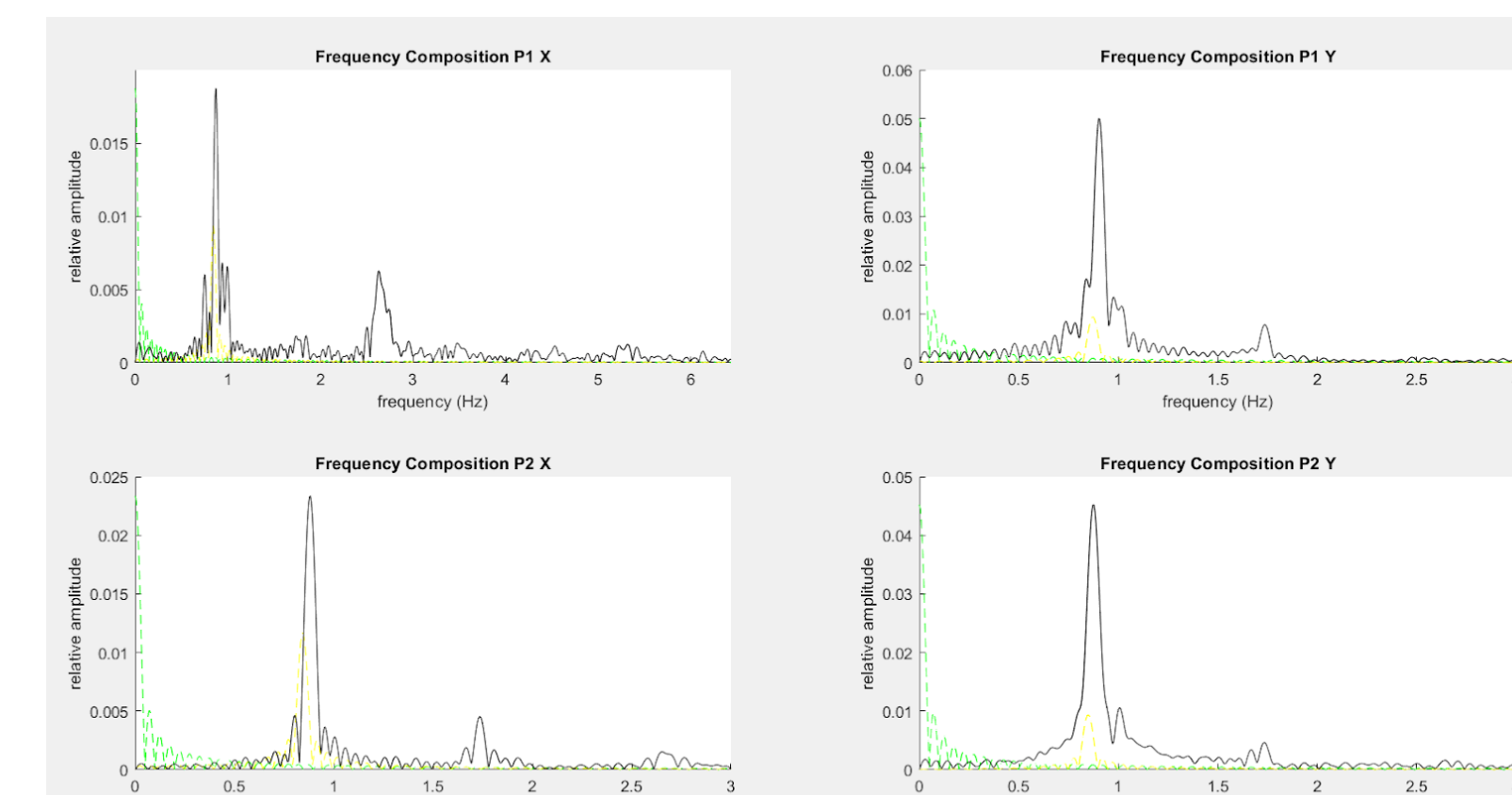
Acceleration in X and Y for Circle Mode (10)



Frequency Components for each Axis

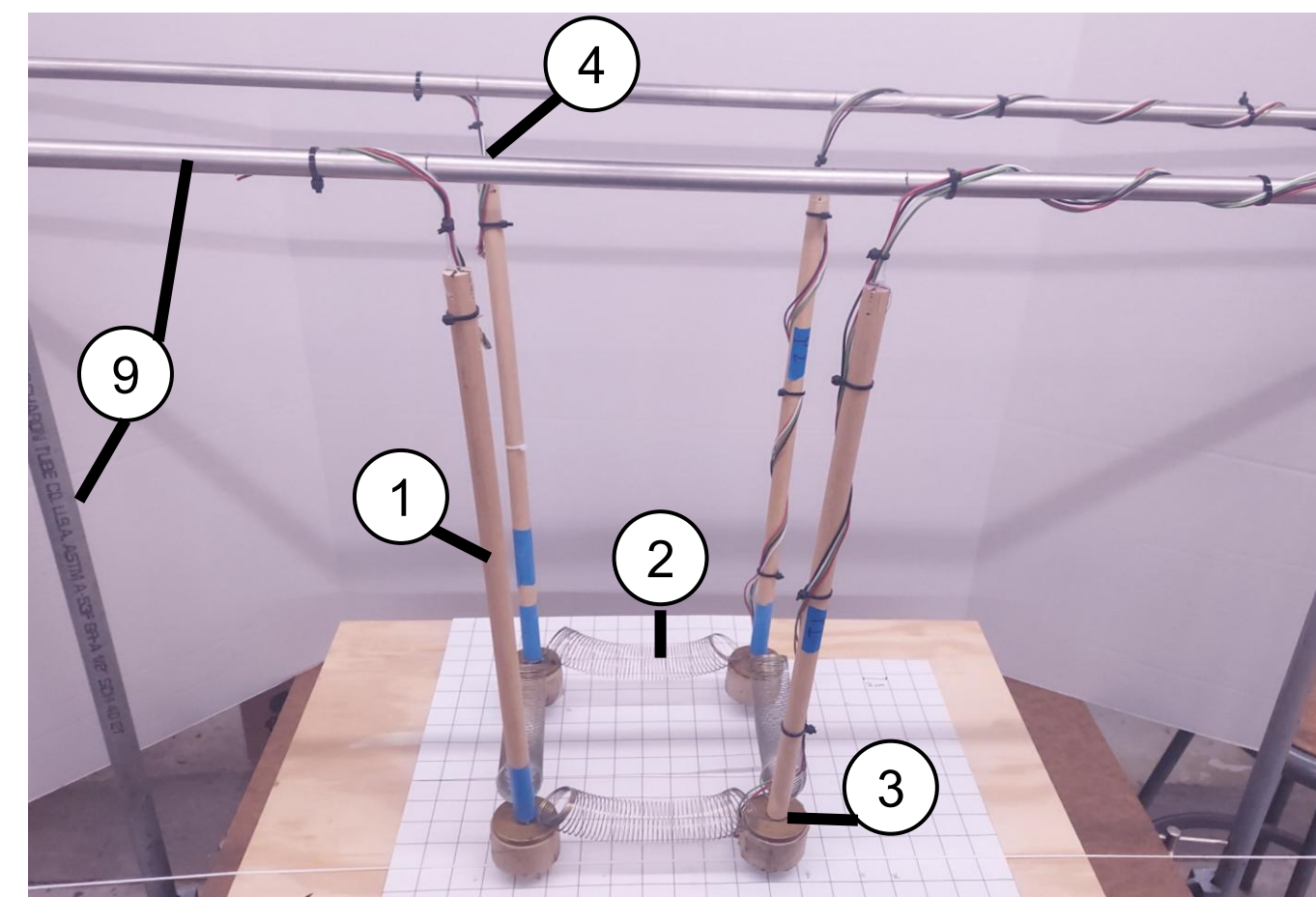


Acceleration in X and Y for Assisted Parallel Rotating Mode (11)

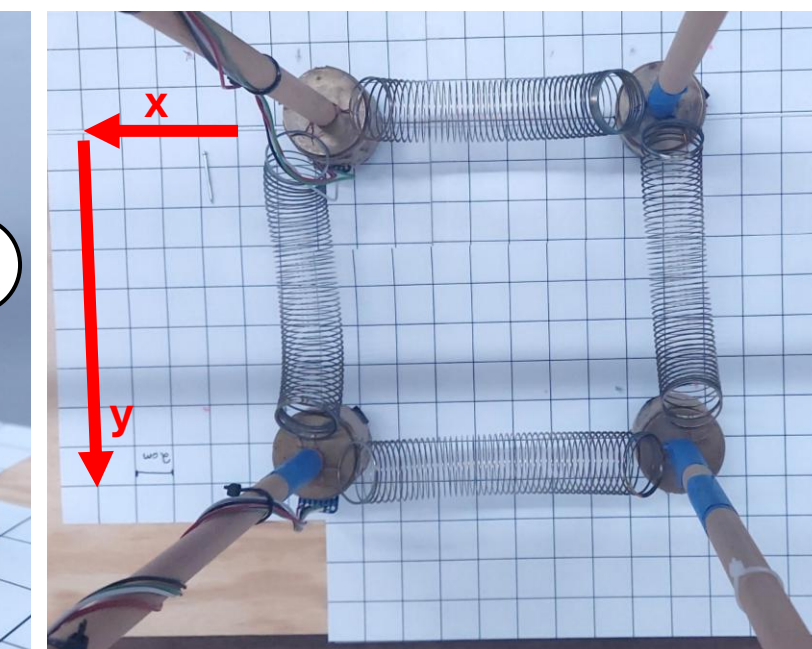
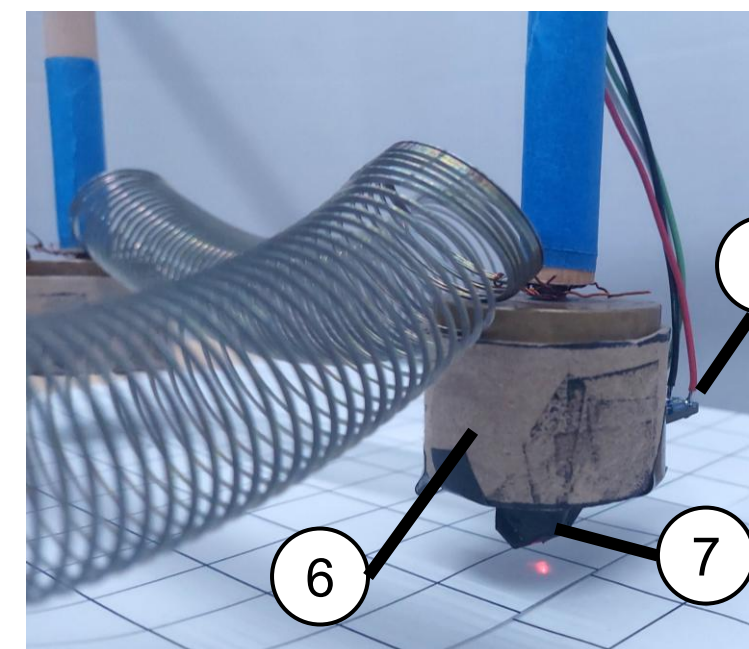
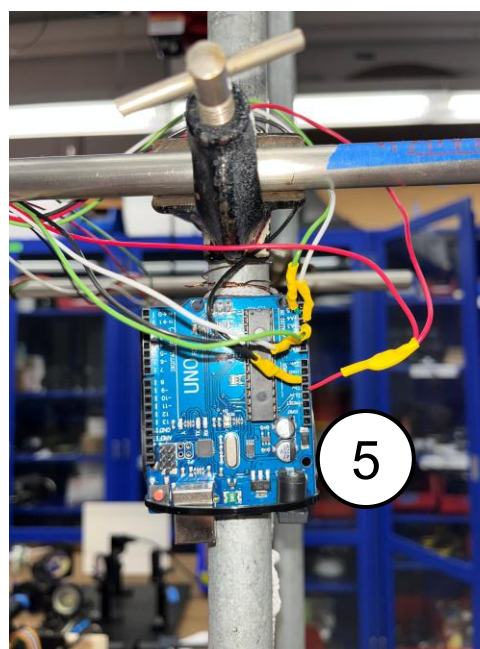


Frequency Components for each Axis

Experimental Setup and Methodology



	Description
1	Wooden Dowel
2	Spring
3	Brass Weights
4	Dummy Wire
5	Arduino Uno R3
6	Cardboard Housing
7	LED tea light
8	Accelerometer
9	Steel Frame



Materials: Four brass weights, four wooden dowels, four identical springs, two ADXL335 accelerometers, an Arduino Uno R3, and four LED tea lights.

Physical Setup: Four vertical steel pipes formed the corners of the frame, spaced 8 inches apart. Two horizontal steel rods were leveled across the pipes, and wooden dowels were suspended from them with fishing line at 8-inch spacing. Brass weights were attached to the dowels, each enclosed in a cardboard housing holding an accelerometer and an LED (taped to leave a single light point). A printed 2-cm grid beneath the apparatus allowed x–y displacement tracking.

Electrical Setup: Accelerometers were mounted inside two of the pendula to measure x and y motion, with wiring routed along the dowels. To avoid asymmetric stiffness, identical dummy wires were added to the remaining pendula. The Arduino read two dual-axis ADXL335 accelerometers at a 500 Hz sample rate, applied oversampling, median filtering, and exponential smoothing, and then used a slow low-pass filter to estimate and remove the gravity component from each axis. The tangential (dynamic) acceleration values in the x and y directions for both sensors was collected at 100 Hz and saved analyzed further with MATLAB.

Conclusion & Discussion

Discussion:

- **Breathing Mode (3):** This mode is a clear example of a normal mode and produced the expected behavior in our experimental setup. Each pendulum's spectrum is well-approximated by a sinc function centered at a nonzero frequency, as expected for a finite-length signal. For comparison, a reference sinc centered at zero is shown in green. Notably, the signal exhibits significant decay.
- **Trapezoid Mode (8):** This mode is non-normal. The acceleration data show periodic energy exchange between pendulums, and the corresponding spectra are far less smooth. The additional peaks arise not only from decay but also from the superposition of the sinusoidal envelope present in the acceleration signal.
- **Circle Mode (10):** Although one might expect this configuration to be normal, both spectral analysis and simulation show otherwise. For motion to require no net work on the springs, each pendulum would need to trace a circle so that the inward force remains constant. Instead, the pendulums attempt to follow the sides of a square, producing forces that are not directed toward the system's center. Consequently, this mode is not normal.
- **Assisted Parallel Rotating Mode (11):** This mode appears non-normal in our trial. However, comparison with simulation reveals that—with the appropriate initial velocity—it is a normal mode. This demonstrates that position alone is insufficient to fully specify a normal mode in 2D.

Conclusions

Our definition of a normal mode—motion of all pendulums at constant frequency in both x and y—adequately characterizes the behaviors observed. The experimental setup successfully captured the expected system dynamics. In two dimensions, initial position alone does not uniquely determine a normal mode. Two anticipated normal modes (10 and 4) proved not to be normal, largely due to small but cumulative energy transfers.